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2015.0 RANGE ROVER (LG), 303-04

FUEL CHARGING AND CONTROLS - TDV8 4.4L DIESEL

DIAGNOSIS AND TESTING

PRINCIPLE OF OPERATION

For a detailed description of the fuel charging and controls system and operation, refer to the relevant Description and Operation section in the Workshop Manual.
REFER to: Fuel Charging and Controls (303-04F Fuel Charging and Controls - TDV8 4.4 L Diesel, Description and Operation).

INSPECTION AND VERIFICATION

WARNING:

Make sure that all suitable safety precautions are observed when carrying out any work on the fuel system. failure to observe this warning may result in personal injury.

CAUTIONS:

- Diagnosis by substitution from a donor vehicle is NOT acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.
- Make sure that absolute cleanliness is observed when working with these components. Always install blanking plugs to any open orifices or lines. failure to follow this instruction may result in damage to the vehicle.

1. Verify the customer concern.

1. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
Low/Contaminated fuel	Fuses
Fuel supply/return line(s)	Glow plug indicator
Fuel tank and filler pipe	Fuel pump module
Fuel leak(s)	Sensor(s)

▪ Fuel filler cap ▪ Fuel filter ▪ Push connect fittings ▪ Fuel rail ▪ Fuel pump ▪ Exhaust Gas Recirculation (EGR) system	▪ Engine Control Module (ECM) ▪ Fuel volume control valve ▪ Fuel pressure control valve ▪ Fuel Rail Pressure (FRP) sensor ▪ Fuel temperature sensor ▪ Fuel injector(s) ▪ EGR system

1. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

1. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Engine cranks, but does not start	▪ Low /Contaminated fuel ▪ Air leakage ▪ Low-pressure fuel system fault ▪ Fuel pump module (lift pump) fault ▪ Blocked fuel filter ▪ Fuel volume regulator blocked /contaminated ▪ Fuel pressure control valve blocked /contaminated ▪ Fuel pump fault ▪ Crankshaft position (CKP) sensor	Check the fuel level and condition. Draw off approximately 1 ltr (2.11 pints) of fuel and allow to stand for 1 minute. Check to make sure there is no separation of the fuel indicating water or other liquid in the fuel. Check the intake air system for leaks. Check the lift pump operation, check the low-pressure fuel system for leaks/damage. Check the fuel filter, check for DTCs indicating a fuel volume or pressure control valve fault. Check the fuel pump. Check the CKP sensor circuits. Refer to the electrical guides.
Difficult to start	▪ Glow plug system fault	Check the glow plug circuits. Refer to the electrical guides. Check the fuel level/condition. Draw off approximately 1 ltr

	(very cold conditions) ▪ Low /Contaminated fuel ▪ Air leakage ▪ Fuel pump module (lift pump) fault ▪ Low-pressure fuel system fault ▪ Blocked fuel filter ▪ Fuel volume control valve blocked /contaminated ▪ Fuel pressure control valve blocked /contaminated ▪ Exhaust gas recirculation (EGR) valve(s) fault	(2.11 pints) of fuel and allow to stand for 1 minute. Check to make sure there is no separation of the fuel indicating water or other liquid in the fuel. Check the intake air system for leaks. Check the lift pump operation, check the low-pressure fuel system for leaks/damage. Check the fuel filter, check for DTCs indicating a fuel volume or pressure control valve fault. Check the EGR system.
Rough idle	▪ Intake air system fault ▪ Low /Contaminated fuel ▪ Low-pressure fuel system fault ▪ Blocked fuel filter ▪ Fuel volume control valve blocked /contaminated ▪ Fuel pressure control valve blocked /contaminated ▪ Exhaust gas recirculation (EGR) valve(s) fault	Check the intake air system for leaks. Check the fuel level /condition. Draw off approximately 1 ltr (2.11 pints) of fuel and allow to stand for 1 minute. Check to make sure there is no separation of the fuel indicating water or other liquid in the fuel. Check the low-pressure fuel system for leaks/damage. Check the fuel filter, check for DTCs indicating a fuel volume or pressure control valve fault. Check the EGR system.
Lack of power when accelerating	▪ Intake air system fault ▪ Restricted exhaust system	Check the intake air system for leakage or restriction. Check for a blockage/restriction in the exhaust system, install new components as necessary. Check for DTCs indicating a fuel pressure fault. Check the EGR system. Check turbocharger actuator.

	- Low fuel pressure - Exhaust gas recirculation (EGR) valve(s) fault - Turbocharger actuator fault	
Engine stops /stalls	- Air leakage - Low /Contaminated fuel - Low-pressure fuel system fault - High pressure fuel leak - Fuel volume control valve blocked /contaminated - Fuel pressure control valve blocked /contaminated - Exhaust gas recirculation (EGR) valve fault	Check the intake air system for leaks. Check the fuel level /condition. Draw off approximately 1 ltr (2.11 pints) of fuel and allow to stand for 1 minute. Check to make sure there is no separation of the fuel indicating water or other liquid in the fuel. Check the fuel system for leaks/damage: Check for DTCs indicating a fuel volume or pressure control valve fault. Check the EGR system.
Engine judders	- Low /Contaminated fuel - Air ingress - Low-pressure fuel system fault - Fuel metering valve blocked /contaminated - Fuel volume control valve blocked /contaminated - Fuel pressure control valve blocked /contaminated - High pressure fuel leak - Fuel pump fault	Check the fuel level/condition. Draw off approximately 1 ltr (2.11 pints) of fuel and allow to stand for 1 minute. Check to make sure there is no separation of the fuel indicating water or other liquid in the fuel. Check the intake air system for leaks. Check the low-pressure fuel system for leaks/damage. Check the high pressure fuel system for leaks, check for DTCs indicating a fuel volume or pressure control valve fault. Check the fuel pump.

Excessive fuel consumption	- Low-pressure fuel system fault - Fuel volume control valve blocked /contaminated - Fuel pressure control valve blocked /contaminated - Fuel temperature sensor leak - High pressure fuel leak - Injector(s) fault - Exhaust gas recirculation (EGR) valve(s) fault	Check the low-pressure fuel system for leaks/damage. Check for DTCs indicating a fuel volume or pressure control valve fault. Check the fuel temperature sensor, fuel pump, etc for leaks. Check for injector DTCs. Check the EGR system.

DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.
REFER to: Diagnostic Trouble Code (DTC) Index - TDV8 4.4L Diesel, DTC: Engine Control Module (ECM) (100-00, Description and Operation).